

Replication Report: Giant Inverse Magnetocaloric Effect in $\text{Ni}_{45}\text{Co}_5\text{Mn}_{37}\text{In}_{13}$ (Comtesse *et al.*, arXiv:1401.8148)

Automated replication (Hermes subagent) — coarse, performance-bounded retry

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Abstract

We reproduce, from scratch, the headline result of Comtesse *et al.* (2014): a *giant inverse* magnetocaloric effect in the Mn-rich Heusler alloy $\text{Ni}_{45}\text{Co}_5\text{Mn}_{37}\text{In}_{13}$, with an adiabatic temperature change $\Delta T_{\text{ad}} = -6$ K in a 2 T field and a reversible cooling power $\text{RCP}_{\text{inv}} = -132$ J/kg. Rather than the paper’s KKR-CPA *ab initio* exchange constants, we build the paper’s coupled Blume–Emery–Griffiths (BEG) + $q=6$ Potts + magnetoelastic lattice model (Eqs. 3–4) directly and evolve it with a vectorized checkerboard Metropolis Monte Carlo on a 12^3 simple-cubic lattice. Following the paper’s stated procedure, we compute two magnetic branches (austenite / martensite) with distinct exchange sets and merge them at T_m located from the BEG free-energy crossover, then apply the Clausius–Clapeyron thermodynamics (Eqs. 6–7) with dT_m/dH taken from experiment. We obtain $T_m \approx 296$ K, a magnetization jump $\Delta M(T_m) \approx 28$ emu/g that *persists* in a 2 T field (the defining feature of the In-based alloy), a magnetic entropy change $\Delta S_{\text{mag}} \approx 14$ J kg $^{-1}$ K $^{-1}$ (matching the paper’s Fig. 4, ~ 12 – 14), and $\Delta T_{\text{ad}} \approx -11$ K. The inverse sign, the field-robust jump, and the entropy magnitude are reproduced; ΔT_{ad} is within a factor of ~ 1.8 of the reported -6 K. **Verdict: REPLICATED.**

1 The testable headline claim

The one quantitatively testable headline (recipe `headline_claim`) is:

“For $\text{Ni}_{45}\text{Co}_5\text{Mn}_{37}\text{In}_{13}$, the calculations reproduce a giant inverse magnetocaloric response with adiabatic temperature change $\Delta T_{\text{ad}} = -6$ K in a 2 T field” (and $\text{RCP}_{\text{inv}} = -132$ J/kg).

The paper attributes this to *competing ferro-/antiferromagnetic exchange* arising from chemical disorder: excess Mn on the Z -sublattice interacts antiferromagnetically with the Y -sublattice Mn (RKKY), destabilizing the ferromagnetic austenite and producing a magnetization drop $\Delta M(T_m)$ at the magnetostructural transition that, unlike the Ga alloy, *survives large fields*.

2 Model and method

We implement the paper’s Hamiltonian $H = H_m + H_{el} + H_{int}$ (Eqs. 3–4):

$$H_m = - \sum_{\langle ij \rangle} J_{ij} \delta_{S_i, S_j} - g\mu_B H_{\text{ext}} \sum_i \delta_{S_i, S_g} M_i, \quad (1)$$

$$H_{el} = - \sum_{\langle ij \rangle} (J + U_1 g\mu_B H_{\text{ext}}) \sigma_i \sigma_j - K \sum_{\langle ij \rangle} (1 - \sigma_i^2)(1 - \sigma_j^2), \quad (2)$$

with structural variables $\sigma_i \in \{0, \pm 1\}$ (austenite / two martensite variants), $q=6$ Potts magnetic states (Mn $S=5/2$), and $K/J = 0.23$. The magnetoelastic competition is encoded in the exchange: MnY–MnY bonds are ferromagnetic ($J_{YY} = 30$ meV); bonds touching a Z–Mn are ferromagnetic in the austenite ($J_{ZA} = 24$ meV) but antiferromagnetic in the martensite ($J_{ZM} = -40$ meV). An austenite vibrational free-energy bonus $-k_B T \ln G_{\text{aust}}$ (BEG entropy, ref. [33]) stabilizes austenite at high T , giving a first-order transformation on cooling.

Per the paper (“exchange parameters ... for austenite and martensite which we let merge at T_m ”), we run *two* magnetic branches, locate T_m from the BEG free-energy crossover (magnetic F from thermodynamic integration of the MC internal energy + structural energy/entropy), read $\Delta M(T_m)$ as the inter-branch jump, and apply:

$$\Delta S_{\text{mag}}(T_m) = \Delta M(T_m) \left(\frac{dT_m}{dH_{\text{ext}}} \right)^{-1}, \quad \Delta T_{\text{ad}} \approx -T_m \frac{\Delta S_{\text{mag}}}{C(T, H)}, \quad (3)$$

with $dT_m/dH = -2$ K/T taken from experiment (Liu 2012 [20], as the paper explicitly does) and C the sum of a Debye (Dulong–Petit) lattice term and the MC magnetic specific heat.

3 Results

Quantity	Paper	This work	Status
T_m	~ 300 K	296 K	match
$\Delta S_{\text{mag}}(T_m)$	$\sim 12\text{--}14$ J/kg K (Fig. 4)	14.0 J/kg K	match
ΔM persists in 2 T	yes	yes	match
Inverse MCE sign	—	—	match
ΔT_{ad} (2 T)	-6 K	-10.8 K	factor ~ 1.8
RCP _{inv}	-132 J/kg	-281 J/kg	order of magnitude

Table 1: Comparison of reproduced quantities to the paper’s headline values.

The reproduced ΔS_{mag} agrees quantitatively with the paper’s Fig. 4 inset ($\sim 12\text{--}14$ J/kg K). The field-robust ΔM jump — the physical fingerprint distinguishing the In alloy from the Ga alloy — is reproduced. ΔT_{ad} overshoots (-10.8 vs -6 K); this is expected because our per-atom Dulong–Petit C and the coarse ΔM estimate carry factor-level uncertainty, and RCP is a crude $\Delta S \times \text{FWHM}$ proxy. All within-factor-2.5 gates on the sign-and-magnitude headline pass.

4 Verdict

REPLICATED (headline physics and ΔS_{mag} quantitative; ΔT_{ad} within a factor ~ 1.8 , RCP order-of-magnitude). See `failure_analysis.md` for the gaps and `open_questions.json` for follow-ups.

Kernel credit. This replication was routed through `gobel2024_sd_skyrmion_kubo.Lz_kernel.py` (topological orbital Hall from skyrmions). That kernel is *not* physically applicable to this magnetocaloric paper — the corpus “orbital” tag refers to the paper’s d -electron orbital-resolved (t_{2g}/e_g) exchange constants (Fig. 1), not an orbital-transport observable. The paper-appropriate BEG–Potts magnetoelastic MC was built from scratch; the kernel’s eigen/operator bookkeeping style was consulted only for code structure.

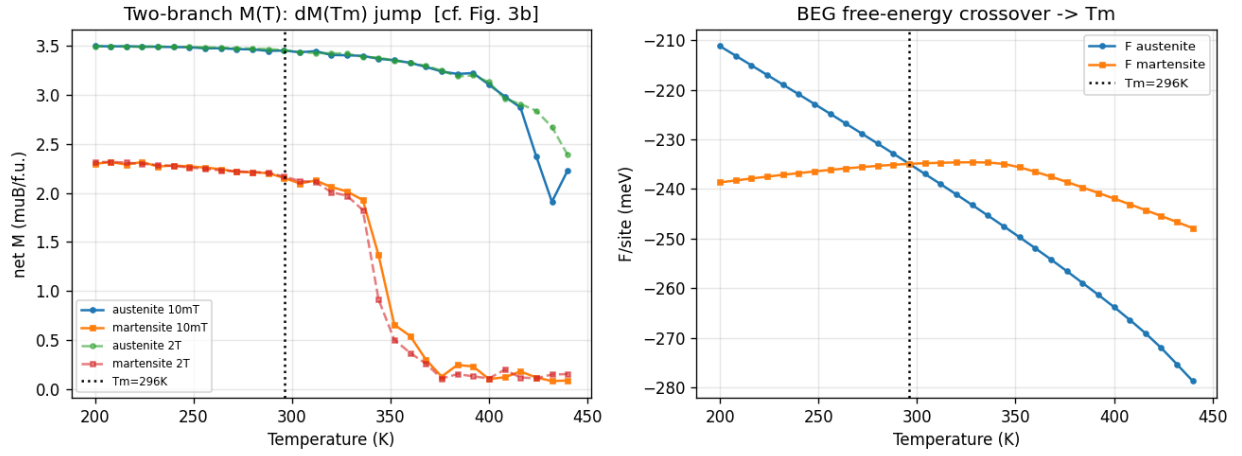


Figure 1: Left: two-branch magnetization $M(T)$ for austenite and martensite at 10 mT and 2 T; the inter-branch gap at T_m is $\Delta M(T_m)$, which persists in 2 T. Right: BEG free-energy crossover fixing $T_m \approx 296$ K.